Exercise: Build a Java-based “Customer Directory” utility (No Spring Boot yet)

Write a simple Java program that:

1. Defines a Customer class
Fields:
int id
String name
String email
boolean active
Includes: Constructor, Getters, toString()

2. Stores a list of customers using ArrayList<Customer>
Initialize a few sample customers manually.

3. Performs the following operations using Collections + Streams

a) Filter active customers
List<Customer> activeCustomers = customers.stream().filter(Customer::isActive).toList();

b) Search customers by name (case-insensitive) — use .filter() on the stream.

c) Sort customers by name — using Comparator.comparing(Customer::getName).

d) Print results cleanly — demonstrate iteration or Streams .forEach().

Customer.java | CustomerRunner.java

```java
package com.sample;

import java.util.ArrayList;

public class Customer {
    int id;
    String name;
    String email;
    boolean active;
    public Customer(int id, String name, String email, boolean active) {
        super();
        this.id = id;
        this.name = name;
        this.email = email;
        this.active = active;
    }
    public int getId() {
        return id;
    }
    public void setId(int id) {
        this.id = id;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public String getEmail() {
        return email;
    }
    public void setEmail(String email) {
        this.email = email;
    }
    public boolean isActive() {
        return active;
    }
    public void setActive(boolean active) {
        this.active = active;
    }
    @Override
    }
    @Override
    public String toString() {
        return "Customer [id=" + id + ", name=" + name + ", email=" + email + ", active=" + active + "]";
    }

}
```

Customer.java | CustomerRunner.java

```java
package com.sample;

import java.util.ArrayList;

public class CustomerRunner {

    public static void main(String[] args) {
        // TODO Auto-generated method stub
        ArrayList<Customer> names=new ArrayList<>();
        names.add(new Customer(1,"Swathi","swathip@gmail.com",true));
        names.add(new Customer(2,"Lahari","laharig@gmail.com",true));
        names.add(new Customer(3,"Kavya","kavyasree@gmail.com",false));
        names.add(new Customer(4,"Chandana","chandanab@gmail.com",false));
        List<Customer> activeCustomers = names.stream().filter(Customer::isActive).toList();
        String searchname="Swathi";
        List<Customer> searchedCustomers=names.stream().filter(c -> c.getName().equalsIgnoreCase(searchname)
        List<Customer> sortedNames=names.stream().sorted(Comparator.comparing(Customer::getName)).toList();
        System.out.println("Active Customer");
        activeCustomers.forEach(System.out::println);
        System.out.println("\nSearched Customer");
        searchedCustomers.forEach(e -> System.out.println(e));
        System.out.println("\nSorted Customers");
        sortedNames.forEach(System.out::println);


    }

}
```

Problems | Declaration | Console | Eclipse IDE for Enterprise Java and Web Developer...

<terminated> CustomerRunner [Java Application] C:\Eclipse-JEE-2025-09\plugins\org.eclipse.justj.openjdk.hot

```
Active Customer
Customer [id=1, name=Swathi, email=swathip@gmail.com, active=true]
Customer [id=2, name=Lahari, email=laharig@gmail.com, active=true]

Searched Customer
Customer [id=1, name=Swathi, email=swathip@gmail.com, active=true]

Sorted Customers
Customer [id=4, name=Chandana, email=chandanab@gmail.com, active=false]
Customer [id=3, name=Kavya, email=kavyasree@gmail.com, active=false]
Customer [id=2, name=Lahari, email=laharig@gmail.com, active=true]
Customer [id=1, name=Swathi, email=swathip@gmail.com, active=true]
```